Solution

Matrix Theory, Quiz 1D — January 23, 2026

For the system of linear equations

$$\begin{aligned} 2x - 2y \phantom{{}-3z} &= 2 \\ 2x + y - 3z &= 8 \\ -2x \phantom{{}+y} + z &= -5 \end{aligned}$$

1C	1B	1A
-6	2	-6
0	11	3
3	-6	2

perform the following:

(a) Write down the augmented matrix. (3)

(b) Use elementary row operations to reduce the matrix of part (a) into reduced row echelon form. (6)

(c) Write the solution to the original system of equations. (1)

(a)
$$\left[\begin{array}{ccc|cccc} 2 & -2 & 0 & 2 & -6 & 2 & -6 \\ 2 & 1 & -3 & 8 & 0 & 11 & 3 \\ -2 & 0 & 1 & -5 & 3 & -6 & 2 \end{array}\right]$$

(b)
$$\xrightarrow[\substack{R_2 - R_1 \to \\ R_3 + R_1 \to}]{\frac{1}{2}R_1 \to} \left[\begin{array}{ccc|cccc} 1 & -1 & 0 & 1 & -3 & 1 & -3 \\ 0 & 3 & -3 & 6 & 6 & 9 & 9 \\ 0 & -2 & 1 & -3 & -3 & -4 & -4 \end{array}\right]$$

$$\xrightarrow{\frac{1}{3}R_2 \to} \left[\begin{array}{ccc|cccc} 1 & -1 & 0 & 1 & -3 & 1 & -3 \\ 0 & 1 & -1 & 2 & 2 & 3 & 3 \\ 0 & -2 & 1 & -3 & -3 & -4 & -4 \end{array}\right] 2$$

$$\xrightarrow[R_3 + 2R_2 \to]{R_1 + R_2 \to} \left[\begin{array}{ccc|cccc} 1 & 0 & -1 & 3 & -1 & 4 & 0 \\ 0 & 1 & -1 & 2 & 2 & 3 & 3 \\ 0 & 0 & -1 & 1 & 1 & 2 & 2 \end{array}\right] -1$$

$$\xrightarrow[\substack{R_2 - R_3 \\ -R_3 \to}]{R_1 - R_3} \left[\begin{array}{ccc|cccc} 1 & 0 & 0 & 2 & -2 & 2 & -2 \\ 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & -1 & -1 & -2 & -2 \end{array}\right]$$

(c)
$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} \begin{pmatrix} -2 \\ 1 \\ -2 \end{pmatrix}$$

1D 1C 1B 1A